

BA0-R systems-modeling prior-art gate closure

R1

Closure date: 2026-08-15

Expected baseline: 7acd2fb297f9e83049409268c18dac629cec5fcd

Closure decision

BA0-R is CLOSED as a bounded prior-art gate. This closure does not define the Base Analysis metamodel and does not authorize BA1 before BA0 responsibility and non-goals are closed.

1 Purpose

BA0-R was opened before Base Analysis semantics and BAE taxonomy design to test whether DDTA was about to reinvent established systems-modeling concepts, and to separate recurring semantic responsibilities from notation-specific or method-specific constructs.

2 Corpus disposition

The BA0-R source records SRC-0039 through SRC-0048 are promoted into the canonical literature registries using the repository's existing verification vocabulary. Detailed access states remain in the source-access registry and in the preserved BA0-R review sidecars.

Source	Status	Note
SRC-0039	verified	NASA-HDBK-1009A; full public text reviewed.
SRC-0040	verified	SysML 2.0; public OMG specification reviewed.
SRC-0041	verified	KerML 1.0; public OMG specification reviewed.
SRC-0042	verified	NASA SE Handbook Rev. 2; full public text reviewed.
SRC-0043	partially_verified	ISO 42010:2022; full normative text not reviewed.
SRC-0044	verified	AADL introduction; full public SEI report reviewed.
SRC-0045	partially_verified	ArchiMate 3.2; licensed normative full text not reviewed.
SRC-0046	verified	Estefan MBSE survey; full public copy reviewed.
SRC-0047	verified	Reflective OPM metamodel; author-uploaded full text reviewed.
SRC-0048	verified	ISO 42010:2011; user-supplied full text reviewed.

The 2011 and 2022 ISO 42010 editions remain separate records. No equivalence of normative clauses is assumed.

3 Closed research conclusions

1. Actor + Asset + Interaction is REJECTED as a sufficient generic systems-modeling core for DDTA.
2. General systems-modeling prior art repeatedly needs semantic responsibilities concerning structure/referents, behavior, information/resources, relations/connections, interfaces/exchanges, state/mode/context, environment/scope, properties/constraints and multiple representations/views.
3. Recurring semantic responsibility does not imply a first-class DDTA metaclass. BA1 must separately decide entity, relationship, role, property/attribute, constraint or derived/projection representation.
4. Actor and Asset are poor generic roots; Interaction is insufficient as a universal relation because mature approaches distinguish connection, interface, flow/transfer, behavior/action and payload semantics.
5. Channel, Store, Contract, Boundary, State, Flow and InformationObject remain OPEN as first-class DDTA types.
6. Generic model-based analysis, views/viewpoints, cross-model consistency/traceability, architecture decisions and rationale are prior art. DDTA novelty must not be claimed at that level.
7. A single canonical underlying repository/model is not a universal prior-art requirement. If DDTA selects a canonical projective Base Analysis, it must justify that choice through DDTA-specific needs such as stable identity, provenance, repeatable analysis, change propagation and re-analysis.
8. View, Model, Projection and Rendering/Diagram are not interchangeable without an explicit DDTA contract.
9. No Chapter 2, 3 or 4 reopen criterion was met.

4 BA0 handoff

Pressure-tested BA0 question

What is the minimum methodology-neutral analyzable representation DDTA needs, grounded in governed documentation and explicit reviewed analytical additions, so that stable system knowledge can be consumed by multiple analysis methods without becoming a second documentation hierarchy or a general-purpose systems-modeling language?

BA0 must explicitly decide:

- whether **Base Analysis** is the correct name;
- what canonical means within the analysis layer without competing with governed-document authority;
- the distinction between grounded fact, derived fact and reviewed analytical addition;
- why DDTA needs a shared/canonical representation rather than only synthetic independent views;
- non-goals excluding SysML replacement, STRIDE-DFD identity, raw NLP/LLM extraction, tool-native authority and threat/finding/control semantics;
- how source inconsistencies are surfaced without silent canonicalization.

5 Deliberately open concepts

No BAE taxonomy is closed. SystemElement, Participant, Capability, Behavior, Interaction, InformationObject, Interface, Connection, Channel, Store/Persistence, Contract, Boundary, State/Mode/Context, Dependency and Allocation remain hypotheses only. TrustBoundary remains especially OPEN because prior-art pressure supports generic scope/responsibility boundaries, not automatic promotion of a security-specific trust classification into the neutral core.

6 Reopen criteria

BA0-R is reopened only if:

1. a newly obtained primary source materially falsifies one of the closed conclusions;
2. the full ISO/IEC/IEEE 42010:2022 text materially contradicts the 2011-based regression conclusion relevant to DDTA;
3. BA0 or BA1 produces a recurring concrete counterexample requiring a missing prior-art check; or
4. a candidate DDTA novelty claim is found to be already satisfied by prior work not represented in BA0-R.

Routine downstream refinement is not a reopen trigger.

7 State after closure

Chapters 2-4	CLOSED / FINAL
Documentation layer	CLOSED
W0	CLOSED
BA0-R systems-modeling prior art	CLOSED
BA0 responsibility/non-goals	CURRENT
BA1 minimal BAE ontology	NOT STARTED

Next authorized microstep

Resume BA0 responsibility/boundary falsification. No BAE type is accepted by this closure.